Introspection Dynamics with Mutation in Additive Games

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Abstract

Cooperation in heterogeneous groups, where individuals differ in resources, productivity, and behavioural responsiveness, underpins collective action across social and biological systems. Introspection dynamics is a learning rule suited to such asymmetric settings: at each time step a randomly selected player compares their current payoff to the payoff an alternative action would have given, and switches with a probability increasing in the difference. We extend introspection dynamics to include *mutation*, where a selected player adopts an action with payoff-independent, player-specific probabilities, together with player-specific selection intensities. From the extended dynamics we rederive and generalise the results of Couto and Pal for *additive* games, those in which the payoff difference a player evaluates when considering a switch is independent of the other players' actions: the stationary distribution remains a product measure, and each player's long-run cooperation probability has an explicit form in terms of their own payoffs, selection intensity, and mutation probabilities alone. We consider the heterogeneous public goods game, where N players may differ in their contributions, public goods multipliers r_i, and selection intensities β_i; the long-run cooperation probability admits a closed form with N terms rather than 2^N. Several structural consequences follow: a player-specific cooperation threshold at $r_i = N$ under symmetric mutation, a neutral-drift regime in which cooperation is governed entirely by mutation bias, and a mutation-selection balance in which aggregate cooperation is affine in the mutation rate, interpolating between selection and neutrality. Mutation also regularises the strong-selection limit $\beta_i \to \infty$, where the mutation-free dynamics degenerate.

1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed, creating a tension between individual incentive and collective benefit [14]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences [10].

A natural learning rule for such settings is *introspection dynamics*, introduced in [3]: at each time step a player compares their current payoff to the payoff they *would have received* had they played an alternative action (keeping all other players fixed), and switches with a probability governed by this difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

Introspection dynamics is closely related to the logit-response dynamics studied in economics [2, 1], in which revising players choose actions with probabilities exponential in their payoffs; the Fermi rule is the two-action logit choice. Asymmetric games have also been approached through evolutionary dynamics directly, for example in [13]. The introspection model of [3, 4], however, contains no mutation: a selected player switches action solely according to the Fermi rule. Mutation, the adoption of a random action independently of payoffs, is a standard ingredient of evolutionary and learning dynamics [7] that accounts for exploration, error, and behavioural noise [15]. In this paper we extend introspection dynamics to include per-player, per-action mutation, together with player-specific selection intensities β_i rather than a single global value.

From the extended dynamics we rederive, and generalise, the additive-game results of [4]. There, the authors extended introspection dynamics to multiplayer asymmetric games: for a general game the transition rate for player i switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension 2^N (more generally L^N when each player has L actions). They identified an important exception: for *additive* games, in which the payoff difference a player evaluates when switching action is independent of the other players' actions (a property also known as *equal gains from switching*), the stationary distribution factorises as a product measure over the players ([4]; Proposition 2), and for the linear public goods game it admits a closed form ([4]; Proposition 3). We show that these results persist under mutation: the stationary distribution remains a product measure, and each player's long-run cooperation probability has the explicit form $p_i = \phi_i(\delta_i)(1 - \mu_{iC} - \mu_{iD}) + \mu_{iC}$.

Specifically, we extend the product-measure result to mutation and to arbitrary finite action sets (Section 3), show that for symmetric rates restricting mutation to the other action is only a reparameterisation of the mutation rate (Section 3.3), record the structural consequences that hold for any additive game (Section 4), specialise them to the heterogeneous public goods game (PGG), for which additivity holds and the closed-form cooperation probability follows (Section 5), and explore the effect of mutation beyond additivity numerically (Section 6). Section 7 concludes.

2 Setup and notation

Consider N ordered players, where each player i selects from a finite action set A_i with $L_i := |A_i| \geq 2$. A state is a profile $\mathbf{a} = (a_1, \ldots, a_N)$, where $a_i \in A_i$ denotes the action currently played by player i, and the state space is $S = \prod_{i=1}^{N} A_i$. Each player i has a payoff function $f_i : S \to \mathbb{R}$ and player-specific selection intensity $\beta_i > 0$. We reserve $b \in A_i$ for a generic alternative action of the player under consideration: we write $\mathbf{a}_{i \to b}$ for the state with player i's action replaced by b, and

$$\Delta f_i(\mathbf{a}; b) := f_i(\mathbf{a}) - f_i(\mathbf{a}_{i \to b})$$

for the *payoff difference* player i evaluates at state $\mathbf{a}$ against the alternative b.

Following [4], a game is *additive* for player i if $\Delta f_i(\mathbf{a}; b)$ depends only on a_i and b, not on $\mathbf{a}_{-i}$, the actions of all individuals other than i; equivalently, the payoff difference a player evaluates when considering a switch is independent of the co-players' actions, a property also known as *equal gains from switching*. Additivity holds precisely when the payoff decomposes as

$$f_i(\mathbf{a}) = g_i(a_i) + h_i(\mathbf{a}_{-i}) \tag{1}$$

for some $g_i : A_i \to \mathbb{R}$ and h_i, in which case $\Delta f_i(\mathbf{a}; b) = g_i(a_i) - g_i(b)$. In words, g_i collects the part of player i's payoff controlled by their own action and h_i the part controlled by the co-players, and the two parts do not interact: switching action changes g_i alone. Indeed, fixing a reference action

$c \in A_i$ and setting $g_i(a_i) := \Delta f_i(\mathbf{a}; c)$ and $h_i(\mathbf{a}_{-i}) := f_i(\mathbf{a}_{i\to c})$ exhibits the decomposition, and the converse is immediate.

Our running special case is that of two actions: $A_i = \{C, D\}$ for all i, coded as $\{1, 0\}$, so that $S = \{0,1\}^N$ with $|S| = 2^N$. Each player then has a single alternative action $\bar{a}_i = 1 - a_i$, and we abbreviate $\Delta f_i(\mathbf{a}) := \Delta f_i(\mathbf{a}; \bar{a}_i)$. Additivity for player i becomes the existence of a constant $\delta_i \in \mathbb{R}$ such that

$$\Delta f_i(\mathbf{a}) = (1 - 2a_i)\,\delta_i, \tag{2}$$

where $\delta_i = g_i(D) - g_i(C) = \Delta f_i\big|_{a_i=0}$ is the payoff difference when player i defects; both cases of (2) follow from $\Delta f_i(\mathbf{a}) = g_i(a_i) - g_i(\bar{a}_i)$.

Note that not all games are *additive*. For example in [3] two of the two-player, two-action games considered to illustrate introspection dynamics are the Prisoner's Dilemma G_1 and the Stag Hunt G_2 of (3), for $0 < c_1, c_2 < b$.

$$G_1 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, b \\ b, -c_2 & 0, 0 \end{pmatrix} \qquad G_2 = \begin{pmatrix} b - c_1, b - c_2 & -c_1, 0 \\ 0, -c_2 & 0, 0 \end{pmatrix} \tag{3}$$

For G_1 we have $\delta_1 = c_1$ and $\delta_2 = c_2$. However for G_2, writing (a_1, a_2) for the state in which player 1 plays a_1 and player 2 plays a_2, $\Delta f_1((1,1)) = b - c_1$ but $\Delta f_1((1,0)) = -c_1$. The Prisoner's Dilemma (as defined by G_1) is *additive* but the Stag Hunt game (as defined by G_2) is not. Further games considered in [3] are the Volunteer's Dilemma and the Volunteer's Timing Dilemma (a generalisation with more than 2 actions), neither of which are *additive*.

Beyond the donation form, the general Prisoner's Dilemma (PD) is parameterised by the reward R, temptation T, sucker payoff S, and punishment P with $T > R > P > S$ and the social-efficiency condition $2R > T + S$:

$$G_3 = \begin{pmatrix} R, R & S, T \\ T, S & P, P \end{pmatrix}.$$

Additivity requires, for each player, the *equality*

$$T - R \;=\; P - S \quad (\text{equivalently } P + R = S + T),$$

that is, the gain from defecting against a cooperator must equal the gain from defecting against a defector. The PD axioms are strict inequalities and do not force this equality; a generic PD therefore fails it. For instance, $T = 5$, $R = 3$, $P = 1$, $S = 0$ satisfies $T > R > P > S$ and $2R = 6 > 5 = T + S$, yet $T - R = 2 \neq 1 = P - S$. The donation form G_1 above is the exceptional case where the equality holds by construction (for player i, both differentials equal c_i).

To define introspection dynamics we introduce the following two components:

- **Fermi function.** Each player i has a personal Fermi function $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$. Since $\beta_i > 0$, the exponential $e^{\beta_i x}$ is strictly increasing in x, so ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$ and $\phi_i(x) + \phi_i(-x) = 1$.
- **Mutation.** Each player i has mutation probabilities $\mu_{ib} > 0$ for each action $b \in A_i$, with $M_i := \sum_{b \in A_i} \mu_{ib} < 1$, where μ_{ib} is the probability that a selected player i adopts action b regardless of the payoff comparison. For two actions the two mutation probabilities are μ_{iC}, towards cooperation, and μ_{iD}, towards defection, so that $M_i = \mu_{iC} + \mu_{iD}$.

At each step, one player i is selected uniformly at random and updates their action in one of two ways. With probability μ_{ib} they mutate to action b, for each $b \in A_i$; the target may be their current action a_i, in which case the state does not change. Otherwise, with the remaining probability $1 - M_i$, they introspect: they draw an alternative action b uniformly from the $L_i - 1$ actions other than a_i and switch to it with probability $\phi_i\big(\Delta f_i(\mathbf{a}; b)\big)$. The transition probability to each state $\mathbf{a}_{i \to b}$ with $b \neq a_i$ is therefore

$$T^{\text{intro}}_{\mathbf{a},\, \mathbf{a}_{i\to b}} = \frac{1}{N}\left[(1 - M_i)\,\frac{\phi_i\big(\Delta f_i(\mathbf{a}; b)\big)}{L_i - 1} + \mu_{ib}\right], \qquad b \neq a_i, \tag{4}$$

and the state is unchanged with the remaining probability,

$$T^{\text{intro}}_{\mathbf{a},\, \mathbf{a}} = 1 - \sum_{i=1}^{N} \sum_{b \neq a_i} T^{\text{intro}}_{\mathbf{a},\, \mathbf{a}_{i\to b}}, \tag{5}$$

which collects the events that the selected player mutates to their current action or introspects without switching. Since ϕ_i is strictly decreasing, a larger payoff gain from keeping the current action yields a smaller switching probability. Setting $\mu_{ib} = 0$ for all possible actions b recovers the introspection dynamics of [3, 4]; the mutation terms add a payoff-independent probability of switching to each action. For two actions the alternative $b = \bar{a}_i$ is the only choice and (4) reads

$$T^{\text{intro}}_{\mathbf{a},\, \mathbf{a}_{i\to \bar{a}_i}} = \frac{1}{N}\big[(1 - \mu_{iC} - \mu_{iD})\phi_i(\Delta f_i(\mathbf{a})) + \mu_{i\bar{a}_i}\big], \tag{6}$$

where the mutation term is indexed by the target action: $\mu_{i\bar{a}_i}$ denotes μ_{iC} when player i currently defects, so that $\bar{a}_i = C$, and μ_{iD} when they cooperate.

Definition 1 (Cooperation probability)**.** *For two actions, given an ergodic chain on S with stationary distribution π, the* cooperation probability *is $p_C = \pi \cdot s$, where $s_{\mathbf{a}} = \frac{1}{N}\sum_i a_i$ is the fraction of cooperators in state $\mathbf{a}$.*

3 Decomposition of the stationary distribution

For additive games without mutation the stationary distribution of introspection dynamics is a product measure, for any finite action sets ([4]; Proposition 2). We show that this structure persists with mutation: for an additive game the next action of a selected player depends only on their current action, so each player evolves as a Markov chain on their own action set, and the joint stationary distribution is the product of the per-player stationary distributions. We prove the general result first, then specialise to two actions, where the marginals take an explicit closed form, and close with an alternative mutation scheme in which a mutating player must change action.

3.1 General action spaces

For each player i, let K_i be the $L_i \times L_i$ stochastic matrix whose entry $K_i(a_i, b)$ is the probability that player i's next action is b given that they are selected whilst playing a_i. In general the switching probabilities in (4) depend on the whole state $\mathbf{a}$ and no such matrix exists; for an additive game $\Delta f_i(\mathbf{a}; b) = g_i(a_i) - g_i(b)$ depends only on a_i and b, so K_i is well defined, with off-diagonal entries

$$K_i(a_i, b) = (1 - M_i)\,\frac{\phi_i\big(g_i(a_i) - g_i(b)\big)}{L_i - 1} + \mu_{ib}, \qquad b \neq a_i, \tag{7}$$

and diagonal entries holding the remaining mass. Every entry of K_i is at least $\mu_{ib} > 0$, so K_i is irreducible and aperiodic and has a unique, strictly positive stationary distribution $\mathbf{m}_i$ on A_i.

Theorem 1 (Stationary distribution)**.** *For any game that is additive for every player, under introspection dynamics with selection intensities $\beta_i > 0$ and mutation probabilities $\mu_{ib} > 0$ with $M_i < 1$, the introspection chain on $S = \prod_i A_i$ has the unique stationary distribution*

$$\pi_{\mathbf{a}} = \prod_{i=1}^{N} \mathbf{m}_i(a_i), \qquad \mathbf{a} \in S, \tag{8}$$

where $\mathbf{m}_i$ is the unique stationary distribution of K_i.

Proof. One step of the chain changes at most one player's action, so the states from which $\mathbf{a}$ is reachable in one step are $\mathbf{a}$ itself and the states $\mathbf{a}_{i\to b}$ with $b \neq a_i$, with transition probabilities

$$T^{\text{intro}}_{\mathbf{a},\mathbf{a}} = \frac{1}{N}\sum_{i=1}^{N} K_i(a_i, a_i), \qquad T^{\text{intro}}_{\mathbf{a}_{i\to b},\mathbf{a}} = \frac{1}{N} K_i(b, a_i).$$

The first expression follows from the holding probability (5) and the diagonal entries $K_i(a_i, a_i) = 1 - \sum_{b\neq a_i} K_i(a_i, b)$. The product form gives $\pi_{\mathbf{a}_{i\to b}}\, \mathbf{m}_i(a_i) = \pi_{\mathbf{a}}\, \mathbf{m}_i(b)$, so

$$\left(\pi T^{\text{intro}}\right)_{\mathbf{a}} = \frac{1}{N}\sum_{i=1}^{N}\Big[\pi_{\mathbf{a}} K_i(a_i, a_i) + \sum_{b\neq a_i} \pi_{\mathbf{a}_{i\to b}} K_i(b, a_i)\Big] = \frac{1}{N}\sum_{i=1}^{N} \frac{\pi_{\mathbf{a}}}{\mathbf{m}_i(a_i)} \sum_{b\in A_i} \mathbf{m}_i(b)\, K_i(b, a_i) = \pi_{\mathbf{a}},$$

since $\sum_b \mathbf{m}_i(b) K_i(b, a_i) = \mathbf{m}_i(a_i)$ by stationarity of $\mathbf{m}_i$. Every switching probability in (4) is positive, so any state can be reached from any other by at most N single-action switches, and every state has a positive holding probability; the chain is therefore irreducible and aperiodic and π is its unique stationary distribution.

□

Theorem 1 reduces the computation of the stationary distribution from one linear system of dimension $\prod_i L_i$ to N systems of dimension L_i.

3.2 Games with two actions

For two actions more is true: the two rows of K_i coincide, and the marginal $\mathbf{m}_i$ takes an explicit closed form. For $L_i \geq 3$ the rows differ in general, since the probability of retaining the current action depends on that action, so the following forgetful property is special to two actions.

Theorem 2 (Individual cooperation probabilities)**.** *For any two-action additive game with constants δ_i, under introspection dynamics with player-specific selection intensity β_i and mutation probabilities $\mu_{iC}, \mu_{iD} > 0$ with $\mu_{iC} + \mu_{iD} < 1$, the long-run probability that player i cooperates is*

$$p_i = \phi_i(\delta_i)(1 - \mu_{iC} - \mu_{iD}) + \mu_{iC}. \tag{9}$$

Proof. We compute the entries of K_i. When $a_i = D$ the payoff difference in (2) is $\Delta f_i = \delta_i$, so by (7) the probability of switching to C when selected is

$$K_i(D, C) = (1 - \mu_{iC} - \mu_{iD})\phi_i(\delta_i) + \mu_{iC} = p_i.$$

When $a_i = C$ the payoff difference is $\Delta f_i = -\delta_i$, so, using $\phi_i(-\delta_i) = 1 - \phi_i(\delta_i)$, the probability of switching to D is

$$K_i(C, D) = (1 - \mu_{iC} - \mu_{iD})\big(1 - \phi_i(\delta_i)\big) + \mu_{iD} = 1 - p_i.$$

The two rows of K_i therefore coincide:

$$K_i = \begin{pmatrix} p_i & 1-p_i \\ p_i & 1-p_i \end{pmatrix},$$

ordering A_i as (C, D). A stochastic matrix with identical rows sends every distribution to that row in a single step, so $\mathbf{m}_i = (p_i,\, 1-p_i)$ is the unique stationary distribution of K_i. Additivity decouples player i from the rest of the population: their action evolves as a two-state Markov chain in its own right, with kernel $\left(1-\frac{1}{N}\right)I + \frac{1}{N}K_i$, which has the same stationary distribution as K_i. Hence the long-run probability that player i cooperates is $\mathbf{m}_i(C) = p_i$.

□

Additivity and the restriction to two actions together make a player's choice forgetful: each time player i is selected they cooperate with the same probability p_i, whatever their current action. This probability is the payoff-driven Fermi value $\phi_i(\delta_i)$, scaled by the chance of no mutation and shifted by the chance of mutating to cooperation.

Theorem 3 (Stationary distribution for two actions)**.** *For any two-action additive game with constants δ_i, the stationary distribution of the introspection chain on $S = \{0,1\}^N$ is the product measure*

$$\pi_{\mathbf{a}} = \prod_{i=1}^{N} p_i^{a_i}(1-p_i)^{1-a_i}, \qquad \mathbf{a} \in S, \tag{10}$$

where p_i is given by (9)*.*

Proof. By the proof of Theorem 2, $\mathbf{m}_i = (p_i,\, 1-p_i)$, so that $\mathbf{m}_i(a_i) = p_i^{a_i}(1-p_i)^{1-a_i}$, and (10) is the product measure (8) of Theorem 1.

□

In the long run the players' actions are therefore independent: the group behaves as N separate biased coins, with player i showing cooperation with probability p_i. Taking the expectation of the cooperator fraction s under this product measure, the cooperation probability is the mean of the individual probabilities,

$$p_C = \frac{1}{N}\sum_{i=1}^{N} p_i. \tag{11}$$

Setting $\mu_{iC} = \mu_{iD} = 0$ in (9) and (10) recovers the two-action case of Proposition 2 of [4]; the argument above shows that the product-measure structure is unaffected by mutation, since mutation alters only the per-player marginal p_i and not the independence between players.

Remark 1 (The two-player donation game)**.** *For the Prisoner's Dilemma G_1 of* (3)*, from [3], the payoff $f_1(\mathbf{a}) = -c_1 a_1 + b a_2$ is additive with $\delta_1 = c_1$, and similarly $\delta_2 = c_2$. Theorems 2 and 3 give the per-player cooperation probability and the product-measure stationary distribution*

$$p_i = \frac{1-\mu_{iC}-\mu_{iD}}{1+e^{\beta_i c_i}} + \mu_{iC}, \qquad \pi_{\mathbf{a}} = \prod_{i=1}^{2} p_i^{a_i}(1-p_i)^{1-a_i}.$$

Setting $\mu_{iC} = \mu_{iD} = 0$ and a common selection intensity $\beta_i = \beta$ recovers $p_i = (1+e^{\beta c_i})^{-1}$ and

$$\pi = \frac{1}{(1+e^{\beta c_1})(1+e^{\beta c_2})}\left(1,\, e^{\beta c_2},\, e^{\beta c_1},\, e^{\beta(c_1+c_2)}\right) \quad \text{over } (CC, CD, DC, DD),$$

the stationary distribution obtained by [3] for this game; they observed that their two-player stationary distribution factorises precisely under the additive condition, of which the donation game is an instance. Mutation preserves the product form while shifting each marginal: with $b = 1$, $c_1 = 0.6$, $c_2 = 0.1$, $\beta = 5$, *and asymmetric mutation* $\mu_{iC} = 0.05$, $\mu_{iD} = 0.15$ *we obtain* $p_1 \approx 0.088$ *and* $p_2 \approx 0.352$, *giving* $\pi_{DD} \approx 0.591$, $\pi_{DC} \approx 0.321$, $\pi_{CD} \approx 0.057$, *and* $\pi_{CC} \approx 0.031$.

3.3 An alternative mutation scheme

In the dynamics of Section 2 a mutating player may adopt any action, including the one they already play. An alternative convention restricts mutation to the other action: for two actions, when player i is selected they mutate to the alternative $\bar{a}_i$ with probability $\nu_{i\bar{a}_i} \in (0,1)$, where, as in (6), ν_{iC} and ν_{iD} are the rates towards each target action, and otherwise apply the Fermi comparison, so that the switching probability of (6) becomes

$$T^{\text{intro}}_{\mathbf{a},\,\mathbf{a}_{i\to\bar{a}_i}} = \frac{1}{N}\big[(1-\nu_{i\bar{a}_i})\phi_i(\Delta f_i(\mathbf{a})) + \nu_{i\bar{a}_i}\big]. \tag{12}$$

For an additive game the selection kernel K_i remains well defined, with

$$K_i(D,C) = (1-\nu_{iC})\phi_i(\delta_i) + \nu_{iC}, \qquad K_i(C,D) = (1-\nu_{iD})\big(1-\phi_i(\delta_i)\big) + \nu_{iD},$$

but its two rows no longer coincide, so the forgetful property in the proof of Theorem 2 fails.

Proposition 1 (Alternative mutation scheme)**.** *For any two-action additive game with constants* δ_i, *under the alternative scheme* (12) *with* $\nu_{iC}, \nu_{iD} \in (0,1)$, *the unique stationary distribution is the product measure* (10) *with marginals*

$$p_i = \frac{(1-\nu_{iC})\phi_i(\delta_i) + \nu_{iC}}{1 + \nu_{iC}\big(1-\phi_i(\delta_i)\big) + \nu_{iD}\,\phi_i(\delta_i)}. \tag{13}$$

Proof. All entries of K_i are positive, so K_i has a unique stationary distribution $\mathbf{m}_i = (p_i, 1-p_i)$, determined by the two-state balance $p_i\,K_i(C,D) = (1-p_i)\,K_i(D,C)$; solving gives $p_i = K_i(D,C)/\big(K_i(D,C) + K_i(C,D)\big)$, which is (13). The proof of Theorem 1 uses only that one player updates at a time and that the selected player's next action depends on their own current action alone; it therefore applies verbatim, and the stationary distribution is the product of the $\mathbf{m}_i$.

□

Proposition 2 (Equivalence of the mutation schemes)**.** *For symmetric rates* $\nu_{iC} = \nu_{iD} = \nu_i$, *the stationary distribution under the alternative scheme coincides with that of the dynamics of Section 2 with symmetric mutation* $\mu_{iC} = \mu_{iD} = \nu_i/(1+\nu_i)$.

Proof. Both stationary distributions are product measures with Bernoulli marginals, so it suffices that the marginals agree. With $\mu_i = \nu_i/(1+\nu_i)$, (9) gives

$$p_i = (1-2\mu_i)\phi_i(\delta_i) + \mu_i = \frac{(1-\nu_i)\phi_i(\delta_i) + \nu_i}{1+\nu_i},$$

which is (13) with $\nu_{iC} = \nu_{iD} = \nu_i$. The map $\nu_i \mapsto \nu_i/(1+\nu_i)$ is a bijection from $(0,1)$ onto $(0,\frac{1}{2})$, so the correspondence runs in both directions.

□

The two chains are different, and in general mix at different speeds, but their long-run behaviour is identical: for symmetric rates, whether a mutating player may re-select their current action is a reparameterisation of the mutation rate, not a modelling choice. Every symmetric-mutation result of Sections 4 and 5 therefore holds for the alternative scheme with μ replaced by $\nu/(1+\nu)$; Figure 3 illustrates the coincidence. For asymmetric rates the equivalence fails: the denominator of (13) depends on $\phi_i(\delta_i)$, so the marginal is no longer affine in the Fermi value, whereas (9) always is, and no fixed choice of μ_{iC}, μ_{iD} reproduces (13) across all payoff differences.

Both conventions appear in the literature. Redrawing from the full action set is the convention of the exploration dynamics of [15], and payoff-independent mutations of this kind drive the equilibrium-selection analyses of adaptive dynamics [12]: a reconsidering player re-samples their options, sometimes landing on the action they already use. Mutation restricted to the other action is the bit-flip mutation of genetic algorithms [11, 8], natural when mutation is read as an error in executing a switch. We emphasise the scope of these results: Propositions 1 and 2 hold for any number of players and for player-specific rates. Since the equivalence acts marginal by marginal, a single population may even mix the conventions, with some players redrawing from the full action set and others restricted to the alternative, and the stationary distribution remains the product of the corresponding marginals. For more than two actions, mutating to a uniformly chosen *other* action still yields a selection kernel that depends only on the player's own action, so Theorem 1 continues to apply there too.

4 Structural consequences

The formula (9) for the individual cooperation probability allows immediate structural conclusions that hold for any two-action additive game, before any specific payoff structure is imposed. We record four: the two limits of the selection intensity, a cooperation threshold, and the effect of mutation on aggregate cooperation. Without mutation, several reduce to properties already observed by [4].

Remark 2 (Neutral drift). *If $\beta_i = 0$ for all i then $\phi_i \equiv \frac{1}{2}$, so the payoff parameters δ_i drop out of* (9) *and cooperation is set entirely by mutation:*

$$p_i = \frac{1 + \mu_{iC} - \mu_{iD}}{2}, \qquad p_C = \frac{1}{2} + \frac{1}{2N} \sum_{i=1}^{N} (\mu_{iC} - \mu_{iD}),$$

so $p_C = \frac{1}{2}$ exactly when $\sum_i (\mu_{iC} - \mu_{iD}) = 0$.

A player insensitive to payoff differences ignores the underlying game: their long-run behaviour is set by their mutation bias alone, and group cooperation reduces to the average asymmetry between the mutation rates.

Proposition 3 (Strong selection). *For each player i,* (9) *confines the cooperation probability to the open interval $(\mu_{iC},\, 1 - \mu_{iD})$, and the endpoints are the strong-selection limits: as $\beta_i \to \infty$,*

$$p_i \to \mu_{iC} \text{ when } \delta_i > 0, \qquad p_i \to 1 - \mu_{iD} \text{ when } \delta_i < 0.$$

Proof. Since $0 < \phi_i(\delta_i) < 1$ and $1 - \mu_{iC} - \mu_{iD} > 0$, (9) gives $\mu_{iC} < p_i < 1 - \mu_{iD}$. As $\beta_i \to \infty$, $\phi_i(\delta_i) = (1 + e^{\beta_i \delta_i})^{-1} \to 0$ when $\delta_i > 0$ and $\to 1$ when $\delta_i < 0$, so $p_i \to \mu_{iC}$ or $p_i \to 1 - \mu_{iD}$ respectively.

□

Without mutation, a unique stationary distribution requires finite β_i [4]: in the limit $\beta_i \to \infty$ the payoff-driven switches become deterministic best responses, the chain is no longer ergodic, and the long-run outcome can depend on the initial state. The strong-selection limit must therefore be taken after computing the stationary distribution at finite selection intensity, as in [3]. For example, in the donation game G_1 of (3) each $\delta_i = c_i > 0$, so as $\beta \to \infty$ the mutation-free stationary distribution collapses to a point mass on the all-defect profile. Mutation removes this degeneracy: the switching probabilities in (6) remain bounded away from 0 and 1, so the chain stays ergodic at any selection intensity and the stationary distribution converges to the non-degenerate product measure with marginals μ_{iC} or $1 - \mu_{iD}$. Mutation therefore extends the exact analysis to arbitrarily strong selection.

Corollary 1 (Cooperation threshold)**.** *Player i cooperates with probability greater than $\frac{1}{2}$ if and only if*

$$\phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{iC}}{1 - \mu_{iC} - \mu_{iD}}.$$

When $\mu_{iC} = \mu_{iD}$ this reduces to $\delta_i < 0$. A sufficient condition for $p_C > \frac{1}{2}$ is that the inequality holds for every i.

Proof. Since $1 - \mu_{iC} - \mu_{iD} > 0$,

$$p_i > \tfrac{1}{2} \iff (1 - \mu_{iC} - \mu_{iD})\phi_i(\delta_i) > \tfrac{1}{2} - \mu_{iC} \iff \phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{iC}}{1 - \mu_{iC} - \mu_{iD}}.$$

When $\mu_{iC} = \mu_{iD}$, the right-hand side equals $\frac{1}{2}$, and since ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$, this is equivalent to $\delta_i < 0$. The claim about p_C follows by averaging (11).

□

With symmetric mutation a player cooperates the majority of the time precisely when the payoff favours cooperation, $\delta_i < 0$; mutation does not move the threshold, it only softens the transition across it. With asymmetric mutation the threshold can be met even when the payoff favours defection. Substituting $\phi_i(\delta_i) = (1 + e^{\beta_i \delta_i})^{-1}$, the condition $p_i > \frac{1}{2}$ is equivalent to

$$\beta_i \delta_i < \ln \frac{\frac{1}{2} - \mu_{iD}}{\frac{1}{2} - \mu_{iC}}.$$

The right-hand side is positive precisely when the mutation is biased towards cooperation ($\mu_{iC} > \mu_{iD}$). Thus when the payoff favours defection ($\delta_i > 0$) a sufficiently strong cooperative bias still yields $p_i > \frac{1}{2}$; conversely, when the payoff favours cooperation ($\delta_i < 0$) a defection bias ($\mu_{iD} > \mu_{iC}$) can drive $p_i < \frac{1}{2}$. The logarithmic form presumes $\mu_{iC}, \mu_{iD} < \frac{1}{2}$. Since $\mu_{iC} + \mu_{iD} < 1$, at most one rate can reach $\frac{1}{2}$, and if one does the threshold question is settled without reference to the payoffs: Proposition 3 confines p_i to $(\mu_{iC}, 1 - \mu_{iD})$, so $\mu_{iC} \geq \frac{1}{2}$ forces $p_i > \frac{1}{2}$ and $\mu_{iD} \geq \frac{1}{2}$ forces $p_i < \frac{1}{2}$, whatever the game.

Proposition 4 (Mutation-selection balance)**.** *Under common symmetric mutation $\mu_{iC} = \mu_{iD} = \mu$ for all i, with $0 < \mu < \frac{1}{2}$, the aggregate cooperation probability is affine in μ,*

$$p_C = (1 - 2\mu)\,\Phi + \mu, \qquad \Phi := \frac{1}{N}\sum_{i=1}^{N} \phi_i(\delta_i),$$

so that $p_C \to \Phi$ as $\mu \to 0$, $p_C \to \frac{1}{2}$ as $\mu \to \frac{1}{2}$, and

$$\frac{\partial p_C}{\partial \mu} = 1 - 2\Phi.$$

Mutation raises aggregate cooperation when $\Phi < \frac{1}{2}$ and lowers it when $\Phi > \frac{1}{2}$.

Proof. With $\mu_{iC} = \mu_{iD} = \mu$, (9) gives $p_i = (1-2\mu)\phi_i(\delta_i) + \mu$. Averaging over i using (11) gives the stated expression for p_C; affinity in μ, the two limits, and the derivative are immediate.

□

Here Φ is the cooperation level of the mutation-free dynamics of [4]. Mutation interpolates linearly between this selection-driven value and neutrality, reaching $p_C = \frac{1}{2}$ as $\mu \to \frac{1}{2}$: noise erases the influence of the payoffs.

5 The heterogeneous public goods game

The *heterogeneous public goods game* [10] is played by N players with player-specific contributions $\alpha_1, \ldots, \alpha_N > 0$ and public goods multipliers $r_1, \ldots, r_N > 1$. The payoff to player i in state $\mathbf{a} = (a_1, \ldots, a_N)$ is

$$f_i(\mathbf{a}) = \frac{1}{N}\sum_{j=1}^{N} r_j \alpha_j a_j - \alpha_i a_i. \tag{14}$$

The first term is a public good shared equally by all players, in which each contribution $\alpha_j a_j$ is scaled by the contributor's multiplier r_j; the second is player i's own contribution cost. We write $P(\mathbf{a}) = \frac{1}{N}\sum_j r_j \alpha_j a_j$ for the shared pool.

The linear public goods game is additive [4], and the heterogeneous version (14), with player-specific multipliers r_i, is additive for the same reason.

Lemma 1 (Additivity of the PGG)**.** *The payoff* (14) *is additive for every player i, with*

$$\delta_i = \alpha_i\left(1 - \frac{r_i}{N}\right), \tag{15}$$

i.e. $\Delta f_i(\mathbf{a}) = (1-2a_i)\,\alpha_i(1 - r_i/N)$, independent of $\mathbf{a}_{-i}$.

Proof. When player i switches from a_i to $\bar{a}_i = 1 - a_i$, only their own term in the pool changes, so $P(\mathbf{a}_{i\to\bar{a}_i}) = P(\mathbf{a}) + \frac{1}{N} r_i \alpha_i (1-2a_i)$. Expanding both payoffs:

$$\begin{aligned} f_i(\mathbf{a}) &= P(\mathbf{a}) - \alpha_i a_i, \\ f_i(\mathbf{a}_{i\to\bar{a}_i}) &= P(\mathbf{a}) + \tfrac{1}{N} r_i \alpha_i (1-2a_i) - \alpha_i(1-a_i). \end{aligned}$$

Subtracting:

$$\begin{aligned} \Delta f_i &= -\frac{r_i}{N}\alpha_i(1-2a_i) + \alpha_i(1-2a_i) \\ &= \alpha_i(1-2a_i)\left(1 - \frac{r_i}{N}\right). \end{aligned}$$

The pool $P(\mathbf{a})$ cancels entirely, confirming additivity with $\delta_i = \alpha_i(1 - r_i/N)$. □

The incentive to switch therefore depends only on the player's own contribution α_i and on whether their multiplier exceeds the group size: $\delta_i < 0$, an incentive to cooperate, precisely when $r_i > N$. The other players' choices never enter. We call player i a *net beneficiary* when $r_i > N$, since each unit they contribute returns more than a unit to them from the pool, and a *net contributor* when $r_i < N$, since each unit they contribute returns less than a unit.

Corollary 2 (Exact cooperation probability)**.** *For the heterogeneous public goods game* (14)*, the long-run cooperation probability is*

$$\boxed{p_C = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1-\mu_{iC}-\mu_{iD}}{1+e^{\beta_i\alpha_i(1-r_i/N)}}+\mu_{iC}\right]}. \tag{16}$$

Proof. By Lemma 1, $\delta_i = \alpha_i(1-r_i/N)$. Substituting $\phi_i(\delta_i) = (1+e^{\beta_i\alpha_i(1-r_i/N)})^{-1}$ into (9) and averaging by (11) gives (16).

□

Aggregate cooperation is thus the average across players of an individual logistic response to their own incentive δ_i, adjusted for mutation; no interaction terms between players survive. Setting $\mu_{iC} = \mu_{iD} = 0$ and taking a single global selection intensity, the product measure (10) reduces to the stationary distribution of the linear public goods game given, for every state, in Proposition 3 of [4], and (16) reduces to the corresponding cooperation level.

Figure 1 validates Corollary 2 against exact values and simulation across group sizes. Table 1 verifies Theorem 3: for $N = 3$ with per-player multipliers $r_1 = 1$, $r_2 = 3 = N$, $r_3 = 9$, contributions $\alpha_1 = 1$, $\alpha_2 = 2$, $\alpha_3 = 3$, $\beta = 2$, and $\mu_{iC} = \mu_{iD} = 0.1$, the individual cooperation probabilities are $p_1 \approx 0.267$, $p_2 = 0.500$, $p_3 \approx 0.900$, and the formula and exact values of $\pi_{\mathbf{a}}$ agree to four decimal places for all eight states.

State $\mathbf{a}$	Formula $\pi_{\mathbf{a}}$	Exact $\pi_{\mathbf{a}}$
DDD	0.0367	0.0367
DDC	0.3299	0.3299
DCD	0.0367	0.0367
DCC	0.3299	0.3299
CDD	0.0133	0.0133
CDC	0.1201	0.1201
CCD	0.0133	0.0133
CCC	0.1201	0.1201

Table 1: Stationary-distribution probabilities $\pi_{\mathbf{a}}$ for all eight states $\mathbf{a} \in \{D, C\}^3$ with $N = 3$, $\alpha_1 = 1$, $\alpha_2 = 2$, $\alpha_3 = 3$, $\beta = 2$, $\mu_{iC} = \mu_{iD} = 0.1$, and per-player multipliers $r_1 = 1<N=r_2=3<r_3=9$. Formula: product measure (10) (Theorem 3); Exact: values from the 2^N linear system obtained using [5]. Both columns are given to four decimal places.

We now specialise the structural consequences of Section 4 to the public goods game, where $\delta_i = \alpha_i(1-r_i/N)$ makes every quantity explicit in the group size and the players' own parameters.

At $\beta_i = 0$ the payoff parameters drop out (Remark 2); the left panel of Figure 2 shows the curves $p_i(\beta_i)$ converging to $p_i = \frac{1}{2}$ at $\beta_i = 0$ before diverging in the direction set by r_i relative to N.

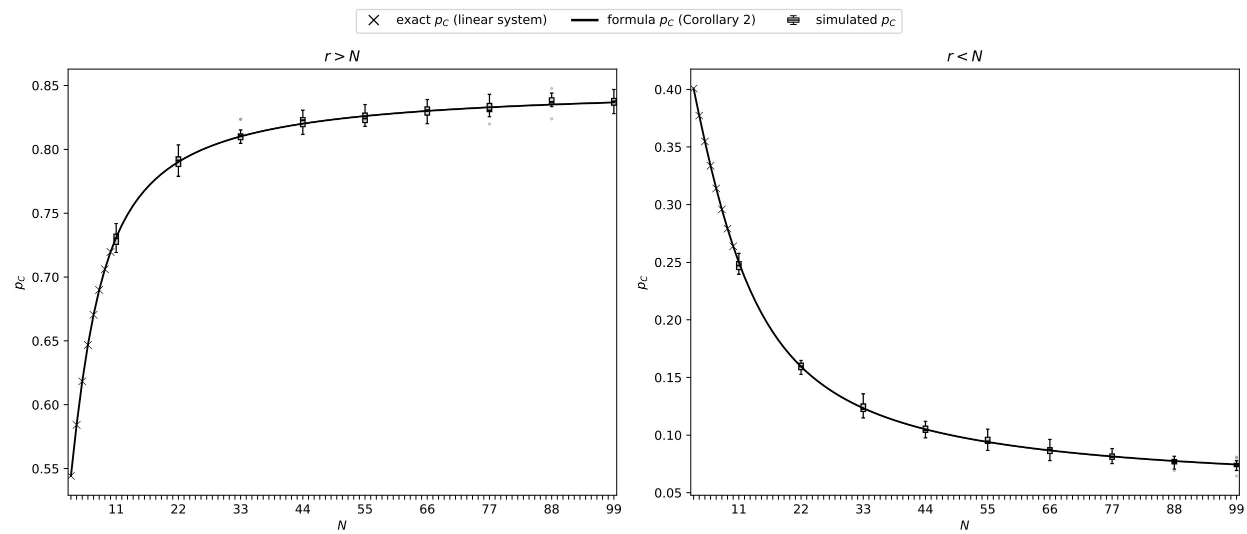

Figure 1: Validation of formula (16) on a heterogeneous group with $\alpha_i = i$, $\beta = 0.5$, $\mu_{iC} = 0.05$, $\mu_{iD} = 0.15$. *Left:* $r = 2N$ (so $r > N$ for all N). *Right:* $r = N/2$ (so $r < N$ for all N). The exact p_C from the full 2^N linear system (feasible only for small N), the formula (16) (Corollary 2), and the distribution of p_C across 19 independent simulation runs of 5,000 steps each (with the first 500 steps discarded as warm-up; outliers shown) are in good agreement. All values were obtained using [5].

For the PGG, $\delta_i < 0$ precisely when $r_i > N$, so the symmetric-mutation threshold of Corollary 1 becomes $r_i > N$: a player cooperates the majority of the time exactly when their personal multiplier on the public good exceeds the group size, that is, when they are a net beneficiary. Without mutation this is the cooperation-dominance condition of [4] (their $c_i < b_i/N$). The centre panel of Figure 2 illustrates this, with curves at $r_i > N$ lying above $p_i = \frac{1}{2}$ and curves at $r_i < N$ lying below it. Since the comparison is between r_i and N, a player with a fixed multiplier becomes a net contributor once the group size exceeds r_i: cooperation in large groups therefore requires multipliers that grow with N. For a net contributor ($r_i < N$, so $\delta_i > 0$) a cooperative mutation bias $\mu_{iC} > \mu_{iD}$ can still raise p_i above $\frac{1}{2}$, as in Corollary 1.

For the PGG, $\phi_i(\delta_i) > \frac{1}{2} \iff r_i > N$, so the sign of $\partial p_C/\partial\mu$ in Proposition 4 is set by whether players are, on average, net beneficiaries: common mutation raises aggregate cooperation when most players have $r_i < N$ and lowers it when most have $r_i > N$. The right panel of Figure 2 shows this mutation-selection balance for a single player: p_i is affine in the symmetric mutation rate μ (Proposition 4), interpolating between the mutation-free value $\phi_i(\delta_i)$ at $\mu = 0$ and $\frac{1}{2}$ at $\mu = \frac{1}{2}$, increasing in μ for a net contributor and decreasing for a net beneficiary.

Figure 3 isolates the role of mutation for a single player. A larger selection intensity β_i makes the player respond more sharply to payoff differences, adopting the higher-payoff action with greater probability, whereas $\beta_i = 0$ is payoff-blind random choice. Without mutation p_i is driven to 0 or 1 as β_i grows, whereas mutation confines p_i to $[\mu_{iC}, 1 - \mu_{iD}]$ (Proposition 3). The two panels show the same player ($r_i = 7$, $\alpha_i = 2$) at $N = 5$ and $N = 200$: increasing the group size alone turns the net beneficiary ($r_i > N$, p_i rising to $1 - \mu$) into a net contributor ($r_i < N$, p_i falling to μ).

Remark 3 (Role of heterogeneity)**.** *For the PGG with $r_i \neq N$, both $\partial p_i/\partial\alpha_i$ and $\partial p_i/\partial\beta_i$ are positive when $r_i > N$ and negative when $r_i < N$: each derivative carries the factor $(1 - r_i/N)$ together with a negative sign stemming from the derivative $\phi_i' < 0$, so its sign is that of $r_i/N - 1$.*

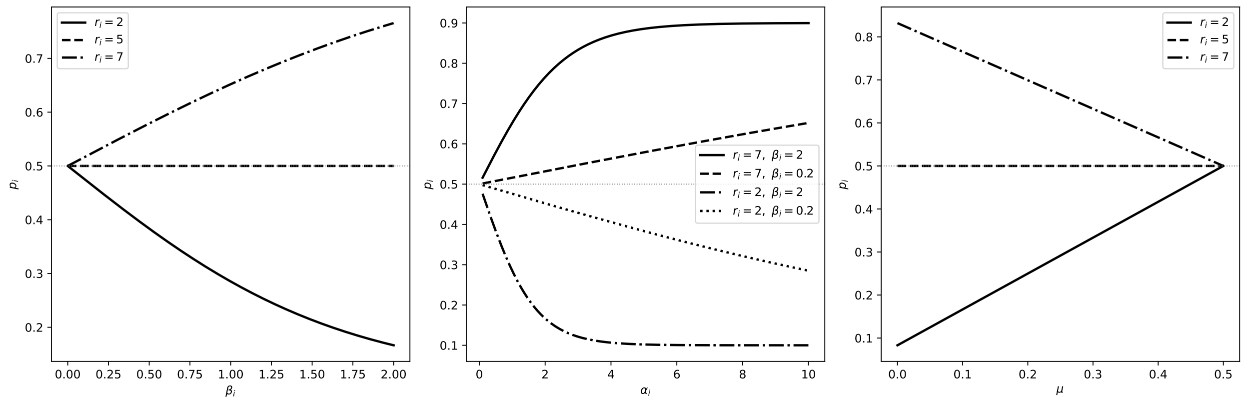

Figure 2: Structural properties of p_i for a single player with $N = 5$ and, unless varied, symmetric mutation $\mu_{iC} = \mu_{iD} = 0.1$. *Left:* p_i as a function of β_i ($\alpha_i = 2$) for $r_i < N$, $r_i = N$, and $r_i > N$; all curves pass through $p_i = \frac{1}{2}$ at $\beta_i = 0$ (Remark 2). *Centre:* p_i as a function of α_i for two values of r_i and β_i; curves above (below) $\frac{1}{2}$ correspond to $r_i > N$ ($r_i < N$). *Right:* p_i as a function of the symmetric mutation rate $\mu_{iC} = \mu_{iD} = \mu$ ($\alpha_i = 2$, $\beta_i = 2$) for $r_i < N$, $r_i = N$, and $r_i > N$; each line is affine in μ (Proposition 4) and reaches $p_i = \frac{1}{2}$ at $\mu = \frac{1}{2}$.

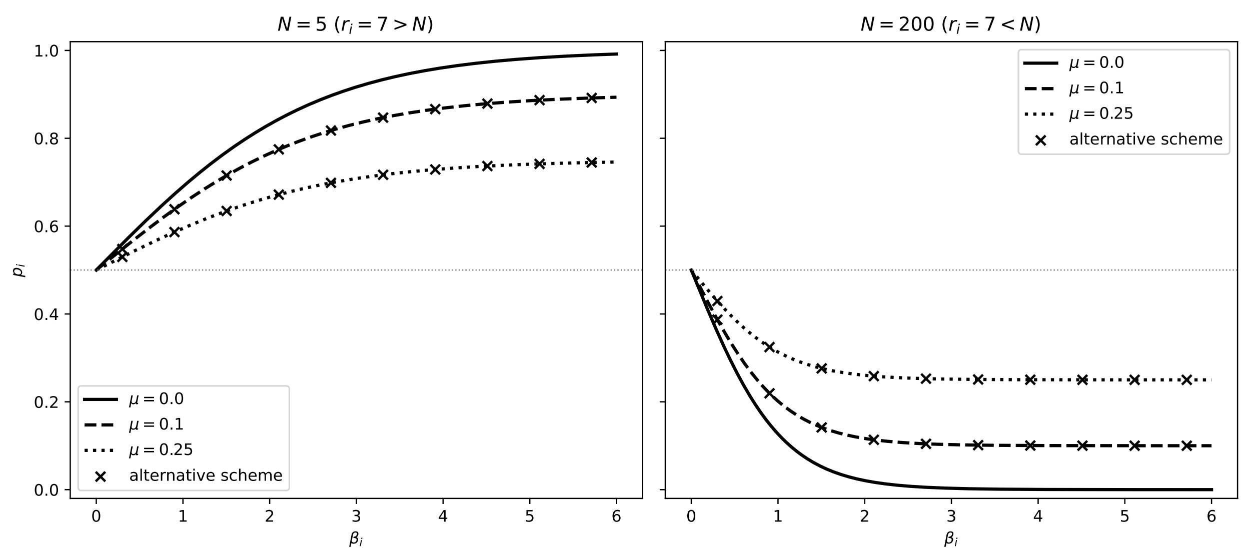

Figure 3: Effect of mutation on the cooperation probability of a single player with multiplier $r_i = 7$ and contribution $\alpha_i = 2$, as a function of the selection intensity β_i for symmetric mutation $\mu \in \{0, 0.1, 0.25\}$, at two group sizes. *Left:* $N = 5$, so $r_i > N$ and the player is a net beneficiary; p_i rises towards $1 - \mu$ as β_i grows. *Right:* $N = 200$, so the same player now has $r_i < N$ and is a net contributor; p_i falls towards μ. In both panels the $\mu = 0$ curve is the mutation-free dynamics of [4], reaching 1 or 0; mutation confines p_i to $[\mu, 1 - \mu]$. The alternative mutation scheme of Section 3.3 with rate $\nu = \mu/(1 - \mu)$ is also shown for $\mu \in \{0.1, 0.25\}$; as established by Proposition 2, it coincides with the corresponding curve. Increasing the group size alone turns the net beneficiary into a net contributor.

Indeed, β_i and α_i enter p_i only through the product $\beta_i\alpha_i$, so a change in selection intensity has the same effect as a proportional change in contribution. Under symmetric mutation, raising the mutation rate instead pulls p_i towards $\frac{1}{2}$, lowering cooperation when $r_i > N$ and raising it when $r_i < N$; with asymmetric mutation p_i is pulled towards its neutral value $(1 + \mu_{iC} - \mu_{iD})/2$.

A net beneficiary ($r_i > N$) cooperates more as they contribute more or respond more strongly to payoff differences; a net contributor ($r_i < N$) does the opposite. Each player's long-run cooperation is thus determined entirely by their own $(\alpha_i, \beta_i, r_i, \mu_{iC}, \mu_{iD})$ and the group size N; the other players' parameters do not enter.

6 Non-additive games

The results so far rely on additivity. When $\Delta f_i(\mathbf{a}; b)$ depends on $\mathbf{a}_{-i}$, the kernel K_i is not defined, the players' actions are coupled, and the stationary distribution must be computed from the full $\prod_i L_i$ linear system. In this section we explore numerically how mutation behaves in that setting, taking the non-linear public goods game as a case study.

In the non-linear PGG the pool is no longer proportional to the total contribution $X(\mathbf{a}) = \sum_j \alpha_j a_j$: each unit contributed changes the marginal value of the next by a factor $w > 0$, so that the payoff to player i is

$$f_i(\mathbf{a}) = \frac{r}{N}\,\frac{w^{X(\mathbf{a})} - 1}{w - 1} - \alpha_i a_i, \tag{17}$$

with the linear game (14) with common multiplier $r_i = r$ recovered in the limit $w \to 1$. For unit contributions ($\alpha_i = 1$) the pool is $(r/N)(1 + w + \cdots + w^{k-1})$ with $k = \sum_j a_j$ cooperators: this is the synergy and discounting game of [9], in which the j-th contribution is worth a factor w of the $(j-1)$-th. This geometric-sum reading requires the total contribution $X(\mathbf{a})$ to be an integer, as it is in both of our examples ($\alpha_i = 1$ here and $\alpha_i = i$ below); for non-integer contributions we regard (17) as the canonical analytic extension of the synergy and discounting model, with the normalisation $w - 1$ retained so that the linear game is recovered as $w \to 1$. For $w > 1$ each additional contribution is worth more than the last: the benefits of cooperation are *synergistically enhanced*, as when collaborators complement one another. For $w < 1$ they are *discounted*, as when the public good saturates and further contributions bring diminishing returns. With unit contributions, switching player i to cooperation adds $(r/N)\, w^{k_{-i}}$ to their share of the pool, where k_{-i} is the number of co-operating co-players, so cooperation is individually favoured precisely when $(r/N)\, w^{k_{-i}} > 1$. For $w = 1$ this is the additive condition $r > N$ of Section 5; for $w \neq 1$ the incentive depends on the co-players and the game is not additive. With discounting ($w < 1$) cooperation is favoured only when few others cooperate; with synergy ($w > 1$) only when enough others do.

Figure 4 reports exact quantities computed from the full 2^N linear system using [5], for $N = 8$. The top row takes unit contributions; the top-left and top-centre panels take $r = 4$, so that $r < N$ and defection dominates the linear game. Three observations follow. First, the $w = 1$ values coincide with the closed form of Corollary 2 and the affine balance of Proposition 4, a numerical check of the additive theory. Second, with discounting the additive pattern persists qualitatively: cooperation is low, mutation raises it towards the neutral level $\frac{1}{2}$, and p_C is no longer exactly affine in μ. Third, with synergy the behaviour changes altogether: cooperation is nearly full at weak mutation, and mutation lowers it towards $\frac{1}{2}$.

The synergy case shows that mutation can produce high cooperation where none is expected. At $w = 1.4$ the game is bistable: all-defect and all-cooperate are both strict equilibria, since the first cooperator loses $1 - r/N > 0$ regardless of w, whilst a defector in a fully cooperative group forgoes $(r/N)w^{N-1} - 1 > 0$. In the strong-selection limit the mutation-free dynamics therefore remain at whichever equilibrium the initial state reaches: started from defection, no cooperation ever arises. Any positive mutation rate makes the chain ergodic, and the top-centre panel of Figure 4 shows the outcome. For $w \leq 1$ strong selection drives cooperation down to the mutation floor μ (Proposition 3 for $w = 1$); for $w = 1.4$ increasing the selection intensity instead drives p_C up, towards approximately $1 - \mu$. Occasional payoff-independent switches allow the group to escape the all-defect equilibrium; once at least three co-players cooperate ($(r/N)w^k > 1$ for $k \geq 3$ at these parameters), introspection favours joining them, and the group is carried to full cooperation. Mutation therefore acts as an equilibrium-selection device in non-additive games, echoing the role of rare mutations in stochastic evolutionary dynamics [7].

Beyond additivity there can also be an optimal, strictly positive mutation rate. For additive games Proposition 4 rules this out: under common symmetric mutation, aggregate cooperation is affine and hence monotone in μ. The top-right panel of Figure 4 shows a smaller multiplier with stronger synergy ($r = 2$, $w = 1.5$), for which cooperation is individually favoured only once four co-players cooperate. At $\beta = 5$ cooperation is nearly full at weak mutation and noise only erodes it. At $\beta = 10$ and $\beta = 20$ the curve is no longer monotone: p_C peaks at $\mu^* \approx 0.02$ and $\mu^* \approx 0.03$ respectively, with the peak falling as selection strengthens. A moderate amount of noise maximises cooperation: enough mutation to seed the escape from the all-defect equilibrium, but not so much as to erode the cooperative one.

Additivity is also precisely the point at which the players' actions decouple. The bottom row of Figure 4 shows the pairwise correlations $\mathrm{corr}(a_i, a_j)$, the Pearson correlation coefficients of the binary actions under the exact stationary distribution (for indicator variables this is the phi coefficient), for the heterogeneous group of Figure 1 (contributions $\alpha_i = i$, $\beta = 0.5$), here with $r = 4$ and symmetric mutation $\mu = 0.05$; since the total contribution reaches 36, mild synergy and discounting ($w = 1.05$ and $w = 0.95$) suffice. For $w = 1$ every pairwise correlation is zero, as the product measure of Theorem 1 requires, despite the heterogeneity. For $w = 0.95$ all pairs are weakly negatively correlated: under discounting a player's incentive to cooperate falls with the contributions of co-operating co-players, favouring anti-coordination. For $w = 1.05$, where the two strict equilibria compete, the correlations are positive and substantial, and increase with both players' contributions: the two largest contributors are the most strongly coupled ($\mathrm{corr}(a_7, a_8) \approx 0.35$). Beyond additivity a player's long-run behaviour is thus tied to their co-players' parameters, in contrast to the additive case, where it depends only on their own; raising the mutation rate erodes every off-diagonal correlation towards zero.

7 Conclusion

Building on the additive-game results of [4], we have shown that introspection dynamics with mutation on any additive game remains exactly solvable: in the long run the players behave independently, so the stationary distribution is a product measure, for any finite number of actions per player (Theorem 1), and its computation reduces to one small linear system per player. The product-measure structure, due to [4] for the no-mutation case, is thus preserved under mutation. For two actions, additivity makes each player's choice forgetful: whenever a player is selected they cooperate with a fixed probability, determined by their own payoff difference, selection intensity, and mutation probabilities alone (Theorem 2), and the stationary distribution is fully explicit (Theorem 3), although this closed form is special to two actions. For the heterogeneous public goods game, the linear payoff structure implies additivity (Lemma 1), and the long-run cooperation probability has a closed form with N terms rather than 2^N (Corollary 2).

The exact analysis does not extend to games that are not additive: when the payoff difference

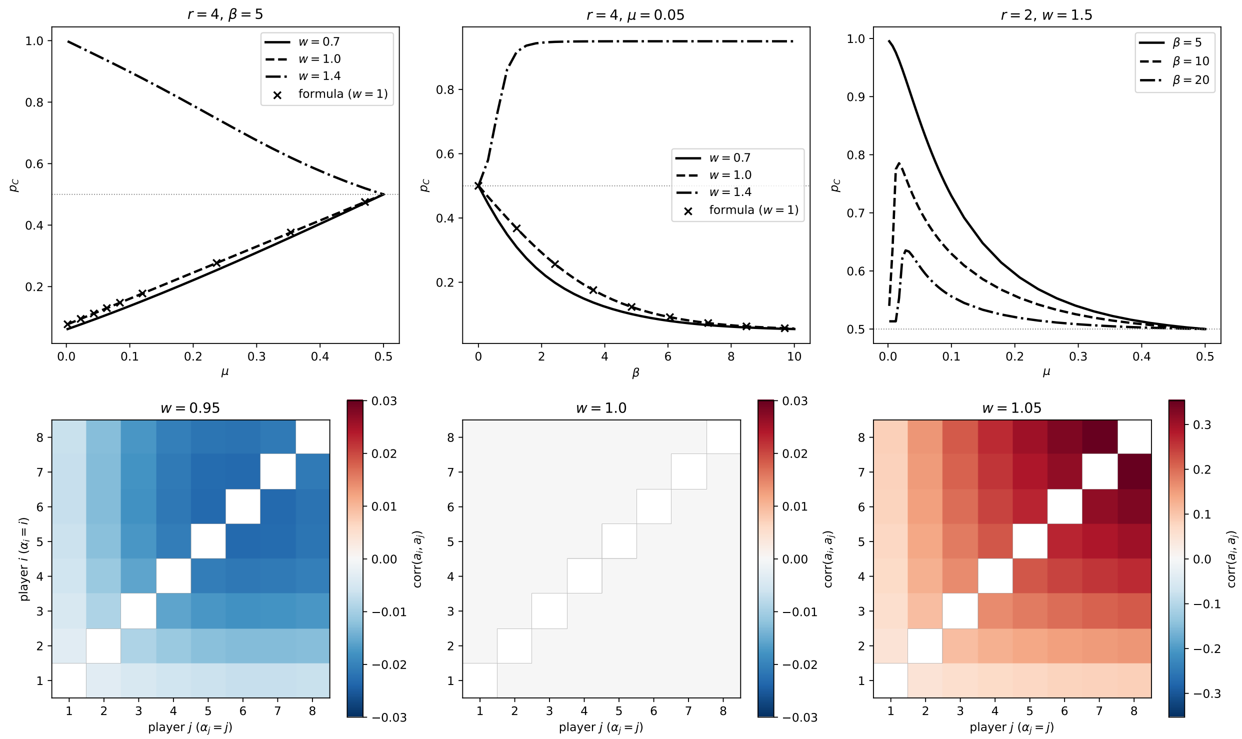

Figure 4: Mutation in the non-linear public goods game (17) with $N = 8$; the top row takes unit contributions ($\alpha_i = 1$). All values were computed exactly from the full 2^N linear system using [5]. *Top left:* p_C as a function of the common symmetric mutation rate μ at $r = 4$, $\beta = 5$, for discounting ($w = 0.7$), the linear game ($w = 1$), and synergy ($w = 1.4$); the closed form of Corollary 2 matches the $w = 1$ values well. The $w = 0.7$ curve follows the additive pattern but is no longer affine, whilst the $w = 1.4$ curve starts near full cooperation and falls towards $\frac{1}{2}$. *Top centre:* p_C as a function of β at $r = 4$, $\mu = 0.05$. For $w \leq 1$ strong selection drives cooperation to the mutation floor μ; for $w = 1.4$ it selects the cooperative equilibrium and p_C approaches $1 - \mu$. *Top right:* p_C as a function of μ for a bistable synergy game ($r = 2$, $w = 1.5$). At $\beta = 10$ and $\beta = 20$ cooperation is maximised at a strictly positive mutation rate, which Proposition 4 rules out for additive games. *Bottom row:* pairwise Pearson correlations $\mathrm{corr}(a_i, a_j)$ of the binary actions under the stationary distribution for the heterogeneous group of Figure 1 (contributions $\alpha_i = i$, $\beta = 0.5$), at $r = 4$ and $\mu = 0.05$. The axes give the player indices i and j, which, since $\alpha_i = i$, also order the players by contribution; diagonals are omitted and the colour scales differ between panels. For $w = 1$ every entry is zero (Theorem 1); for $w = 0.95$ all pairs are weakly negatively correlated; for $w = 1.05$ the correlations are positive and increase with both players' contributions.

a player evaluates depends on the co-players' actions, the players' behaviours are coupled, the stationary distribution is no longer a product measure, and the full 2^N linear system must be solved in general. Games such as the Stag Hunt contain payoff cross terms that cannot be eliminated by rescaling, so no reformulation brings them within the scope of the closed forms. Our numerical exploration of the non-linear public goods game (Section 6) shows that mutation then takes on further roles: it selects among coexisting strict equilibria, cooperation can be maximised at a strictly positive mutation rate, and the players' actions become correlated. Characterising this behaviour analytically remains open.

Throughout, the mutation rates are exogenous: they model trembles, experimentation, and behavioural noise in the players' decisions. On this reading our results describe how noisy self-reflection shapes long-run behaviour: noise keeps every action in play, confines each player strictly inside the pure conventions however strong selection becomes (Proposition 3), and leaves cooperation at a mutation-selection balance between the payoff-driven Fermi level and the neutral $\frac{1}{2}$ (Proposition 4). Whether such noise would itself be selected for, if update rules were heritable and fitness were the expected stationary payoff [15], is a separate question that the closed forms make tractable, and a natural direction for future work.

The source code, figures, and data for this paper are under version control and available at [6], in line with best practices for research software [16].

8 Acknowledgements

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